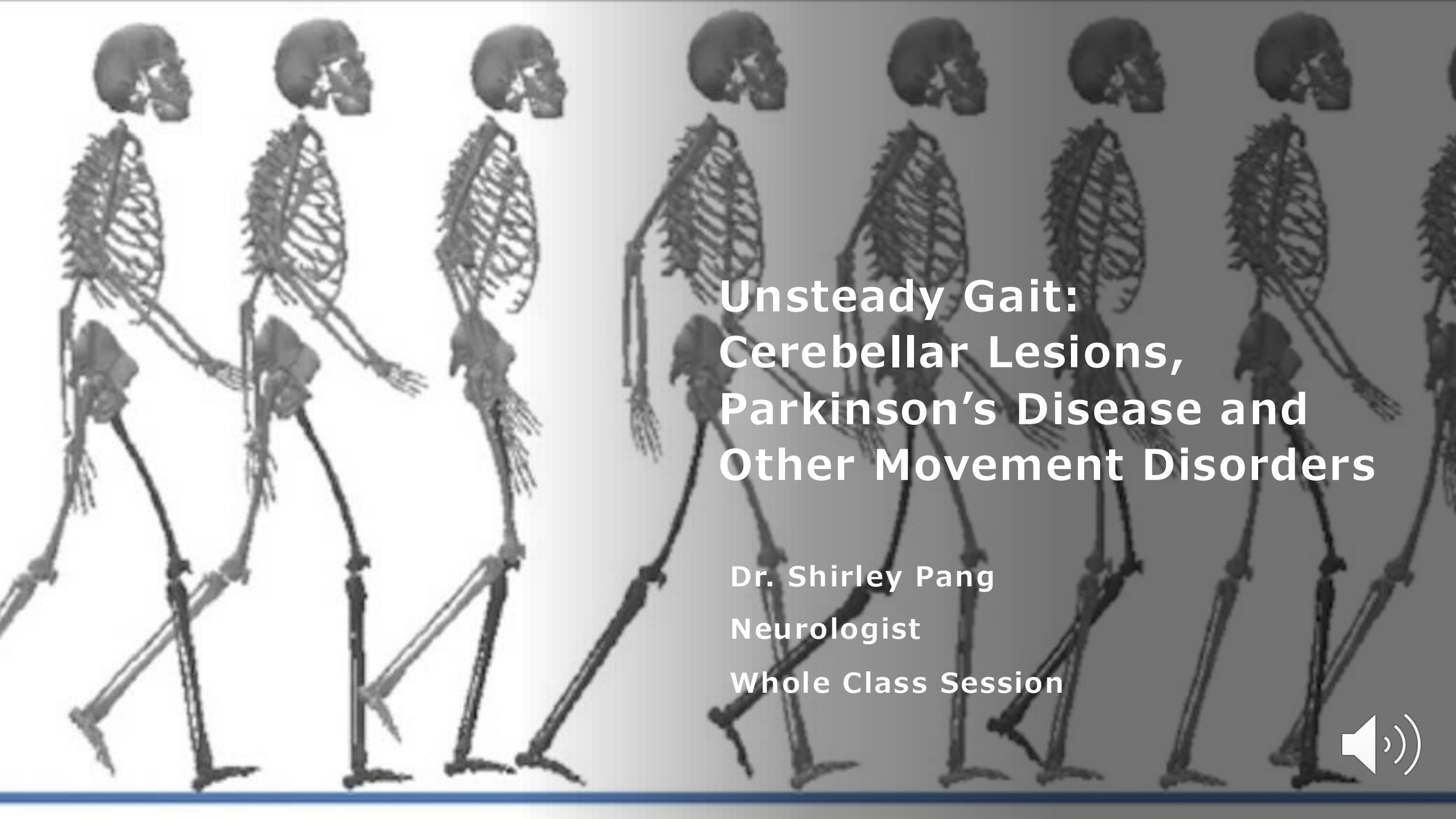




Unsteady Gait: Cerebellar Lesions, Parkinson's Disease and Other Movement Disorders

Dr. Shirley Pang

Neurologist

Whole Class Session



Gait

Complex interplay of many parts of the nervous, musculoskeletal and cardiorespiratory systems

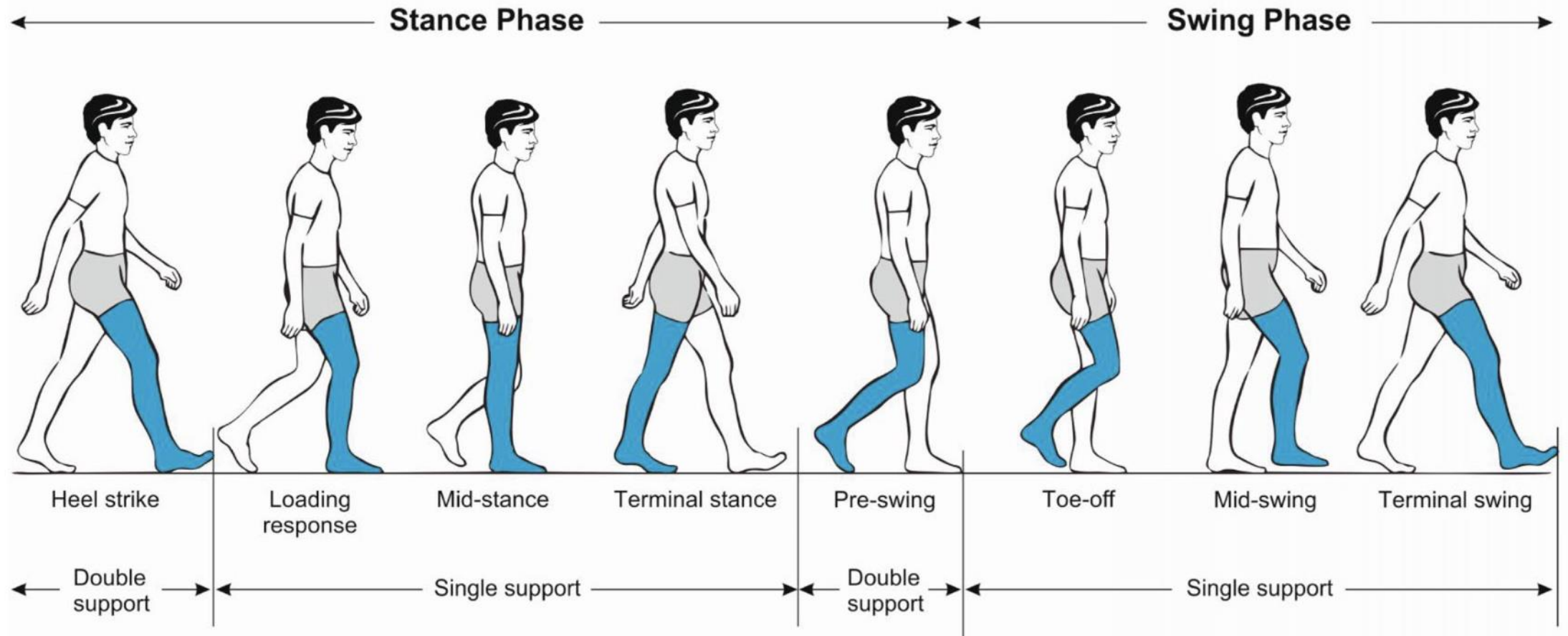
Influenced by age, personality, mood and sociocultural factors

A sensitive indicator of overall health status

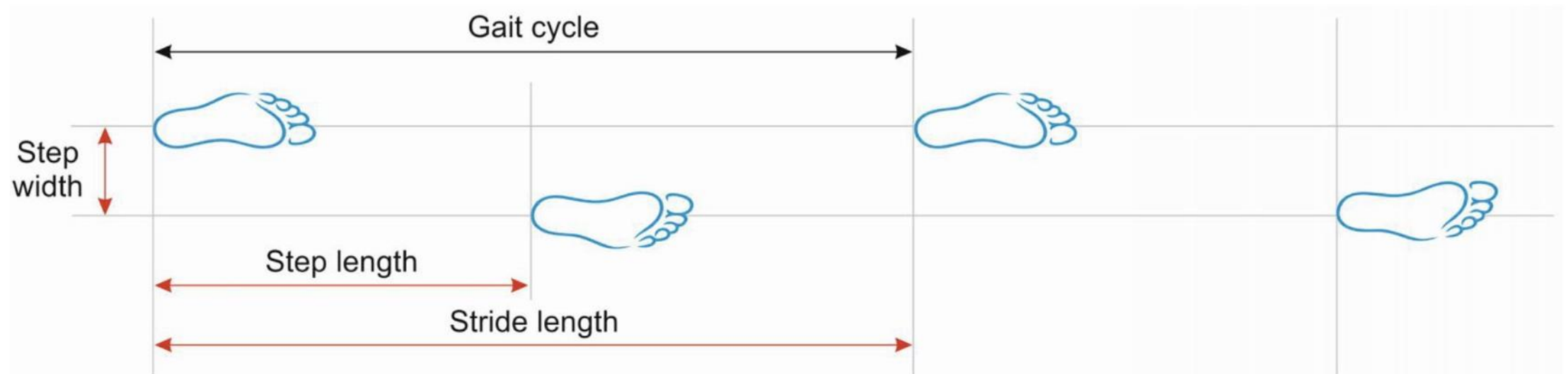
Self-selected walking speed closely correlates with life expectancy in elderly persons



Gait cycle



Gait measurements



Pirker & Katzenschlager, Wien Klin Wochenschr 2017;129:81-95

- Walking speed (m/s)
- Cadence (steps/min)
- Step width (measured from midpoint to midpoint of both heels)
- Step length (Measured from the point of foot contact to the point of contralateral foot contact)
- Stride length (linear distance covered by one gait cycle)



Causes of gait disorders



NEUROLOGICAL
CONDITIONS



ORTHOPEDIC
CONDITIONS

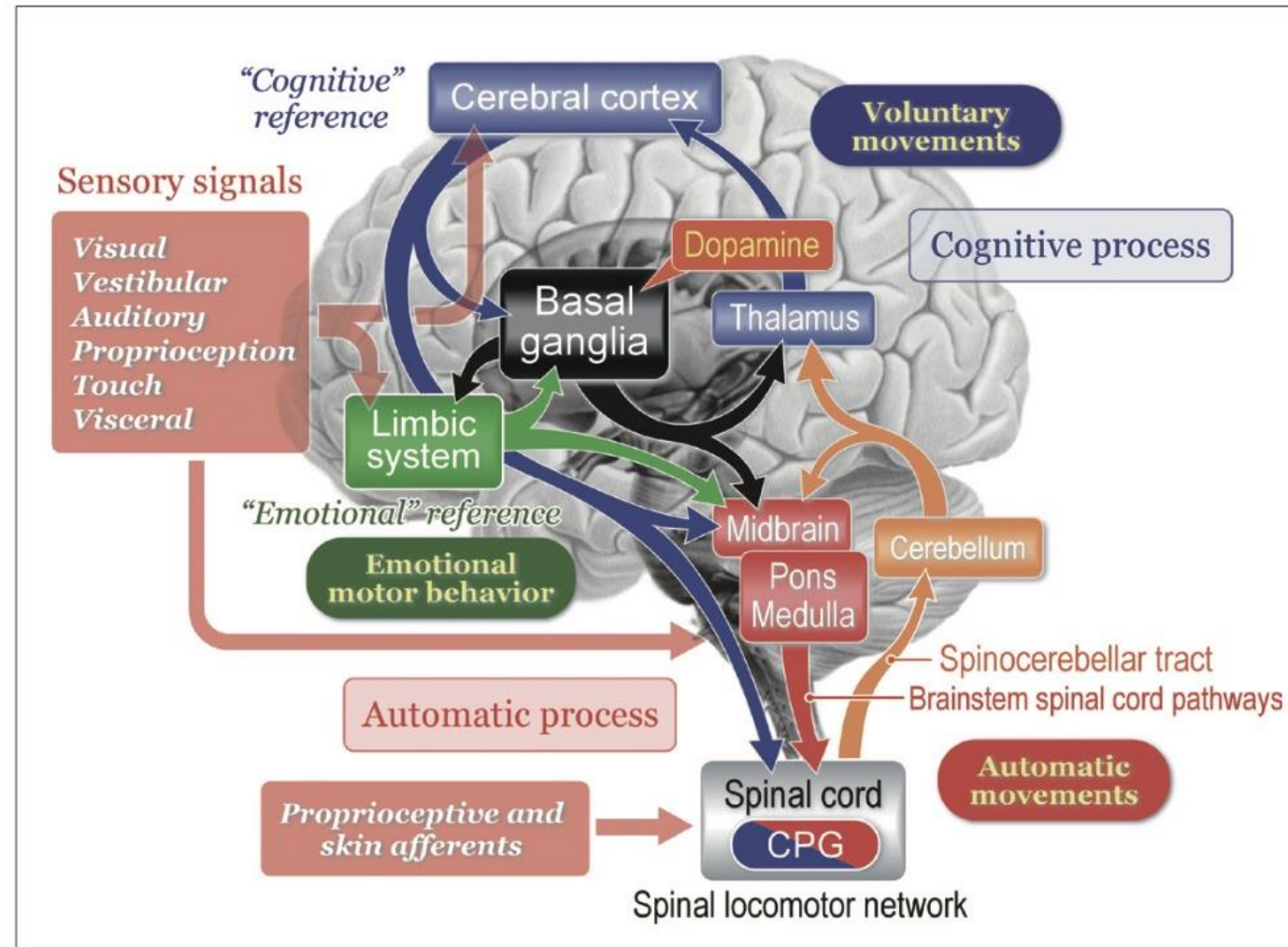


MEDICAL CONDITIONS

- Prevalence of gait disorders increases with age.
 - 10% in people of age 60-69
- 60% in community-dwelling people over 80 years



Physiological basis of gait



Common neurological causes of gait disorder

Sensory
ataxia

Parkinsonism

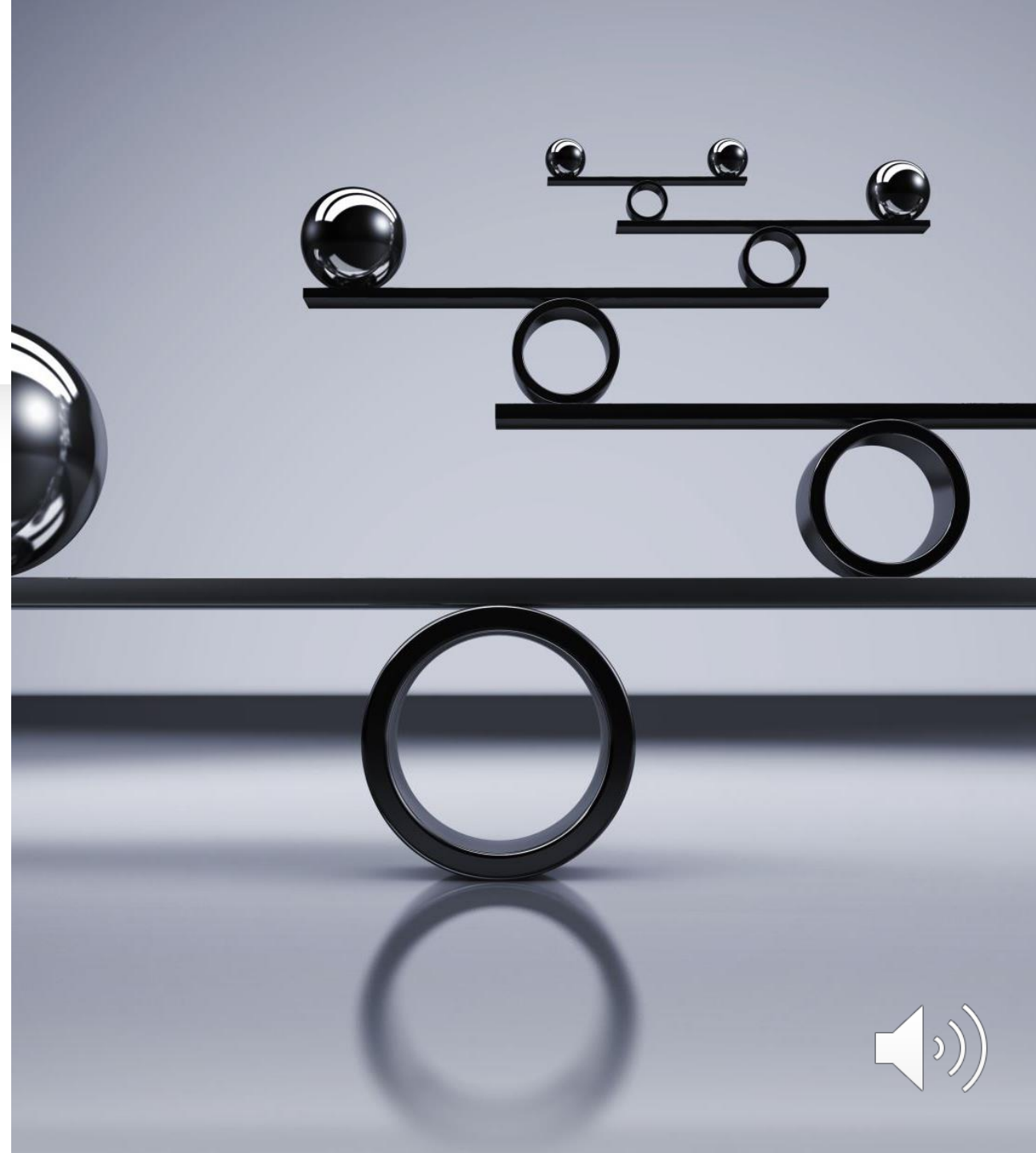
Frontal gait
disorder

Cerebellar
ataxic gait
disorder



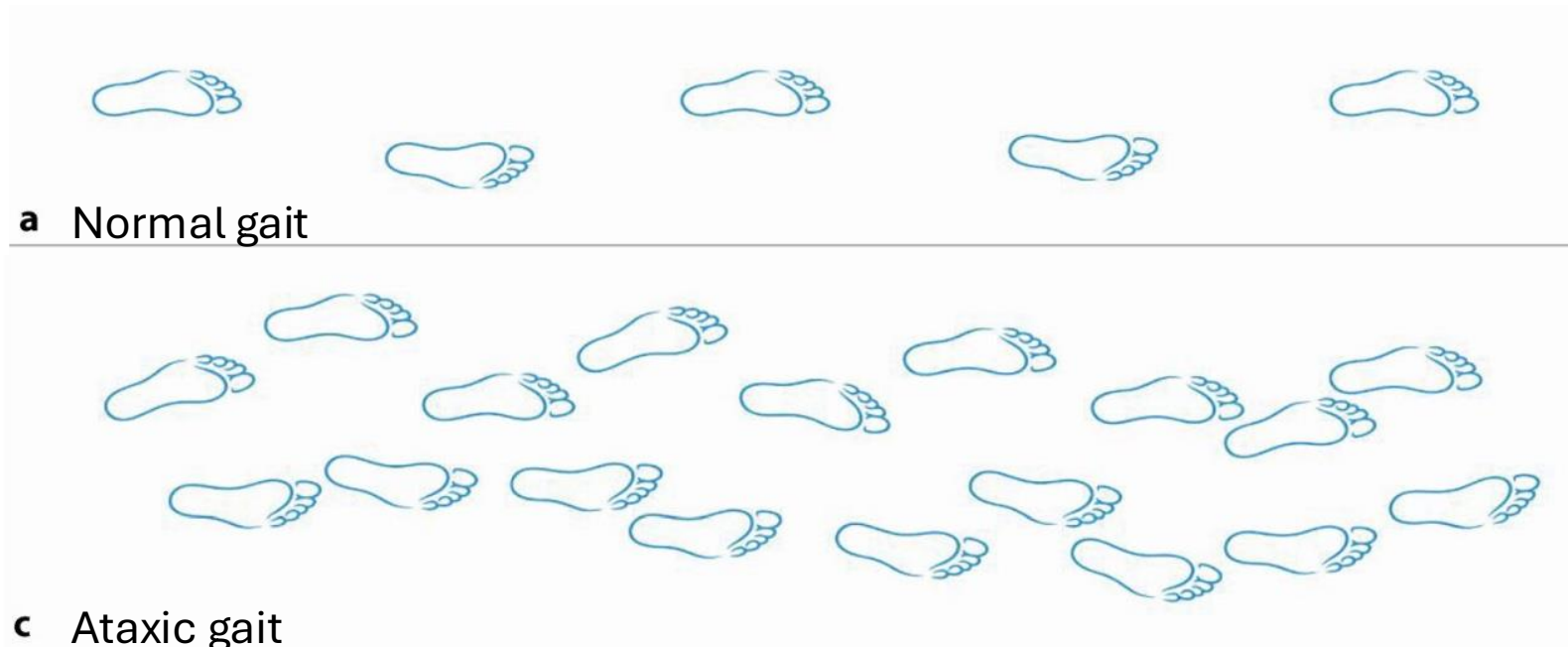
Cerebellum and gait

- Cerebellum important in the act of balancing and coordinating movement
- Cerebellar ataxia: poor coordination of gait and balance, limb and eye movement



Gait ataxia

- Amongst the first cerebellar deficits in cerebellar disorders
- Disabling symptom with a high risk of falls
- “Drunken gait”



Tandem gait



Tandem gait
in cerebellar
ataxia



Video taken from onlinemedicalvideo YouTube Channel

Causes of cerebellar ataxia

Vascular

Toxic (e.g. alcohol, drugs such as anti-convulsants)

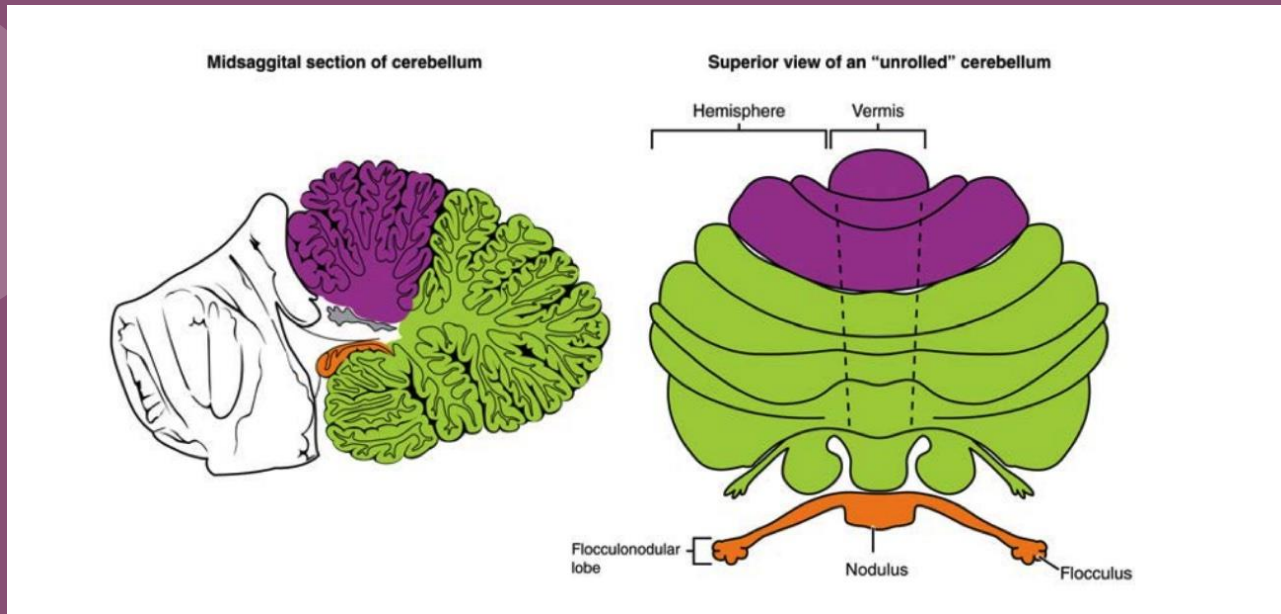
Degenerative (e.g. hereditary cerebellar ataxias)

Inflammatory (e.g. multiple sclerosis)

Neoplastic

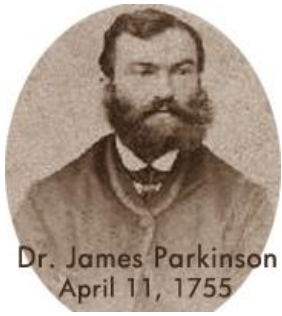


Cerebellar lesions and gait



- Control of balance is predominantly localized in the vermis of the cerebellum
- Paravermal structures are important for the control of the timing and amplitude of targeted limb movements
- Unilateral lesions of cerebellar hemispheres induce ipsilateral limb ataxia
- Isolated lesions of the vermis or the paravermal structures may cause gait ataxia without affecting the limbs





Parkinson's disease

- Shaking palsy:
- “Involuntary tremulous motion with lessened muscular power, in parts not in action even when supported, with a propensity to bend the trunk forward and to pass from a walking to a running pace”
- Other features:
 - Unilateral onset
 - Slow progression
 - Non-motor symptoms such as constipation, drooling of saliva, sleepiness, delirium

AN
ESSAY
ON THE
SHAKING PALSY.

BY
JAMES PARKINSON,
MEMBER OF THE ROYAL COLLEGE OF SURGEONS.

LONDON:
PRINTED BY WHITTINGHAM AND ROWLAND,
Goswell Street,
FOR SHERWOOD, NEELY, AND JONES,
PATERNOSTER ROW.
1817.

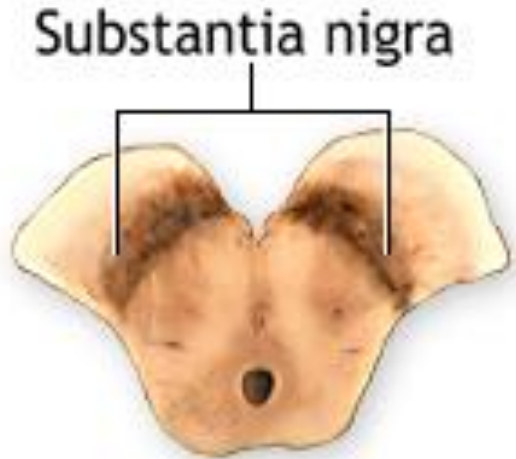


PD: Pathology

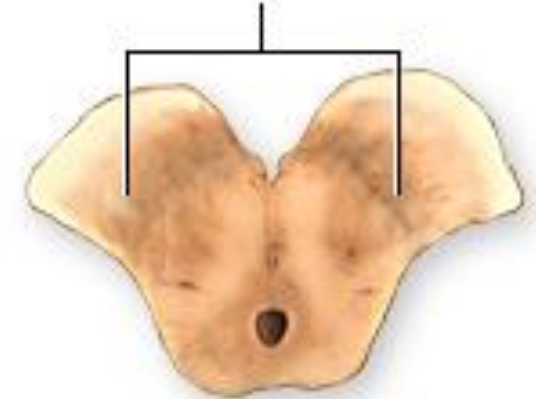
- Loss of dopaminergic neurons in the substantia nigra pars compacta



Cut section
of the midbrain
where a portion
of the substantia
nigra is visible

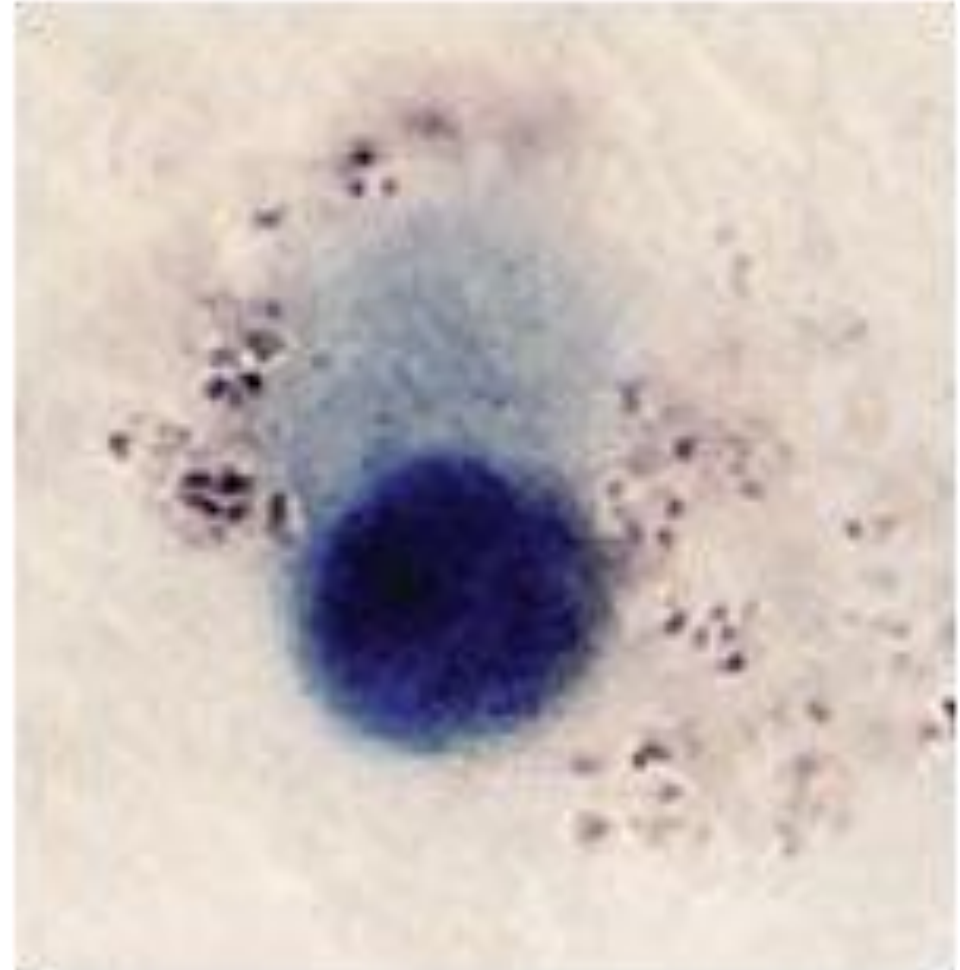


Diminished substantia
nigra as seen in
Parkinson's disease

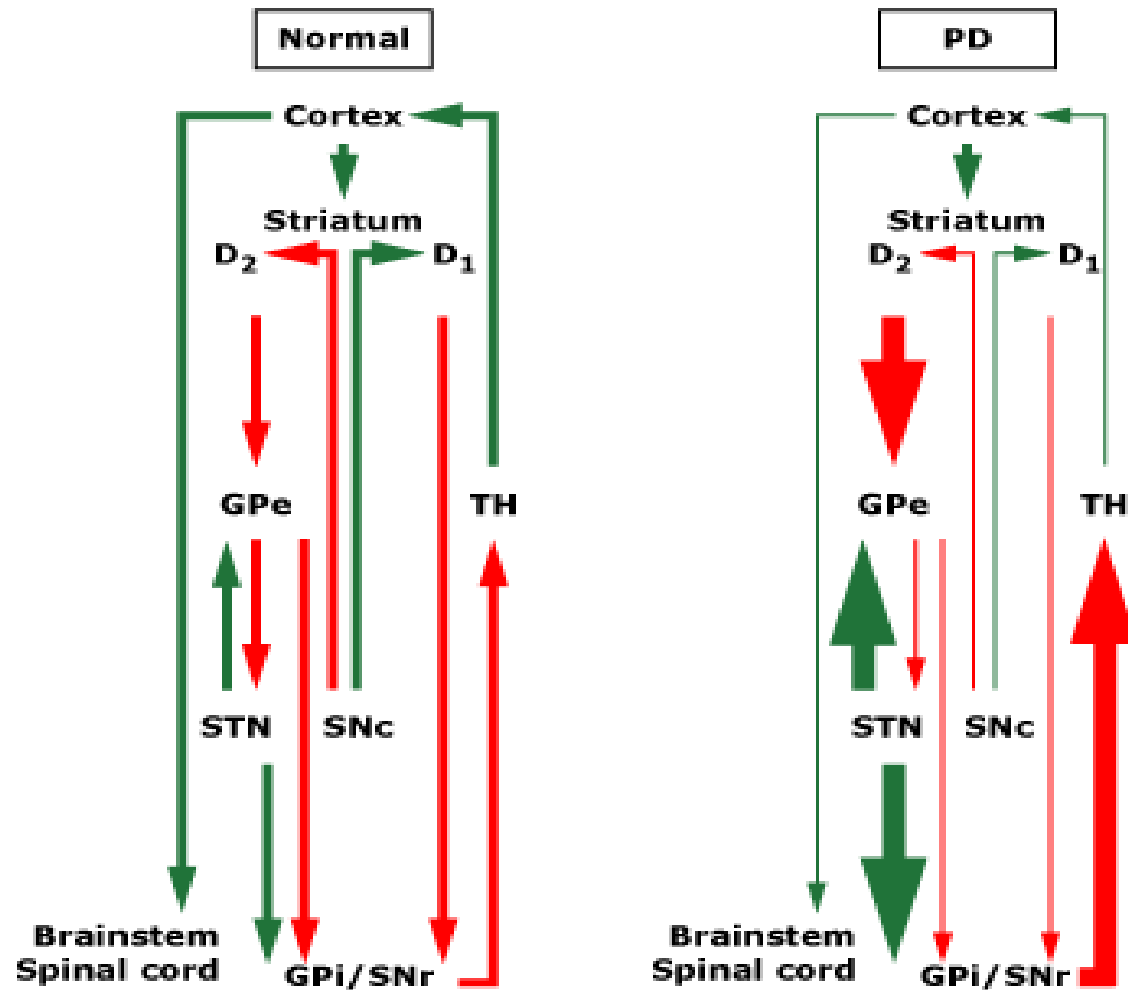


PD: Pathology

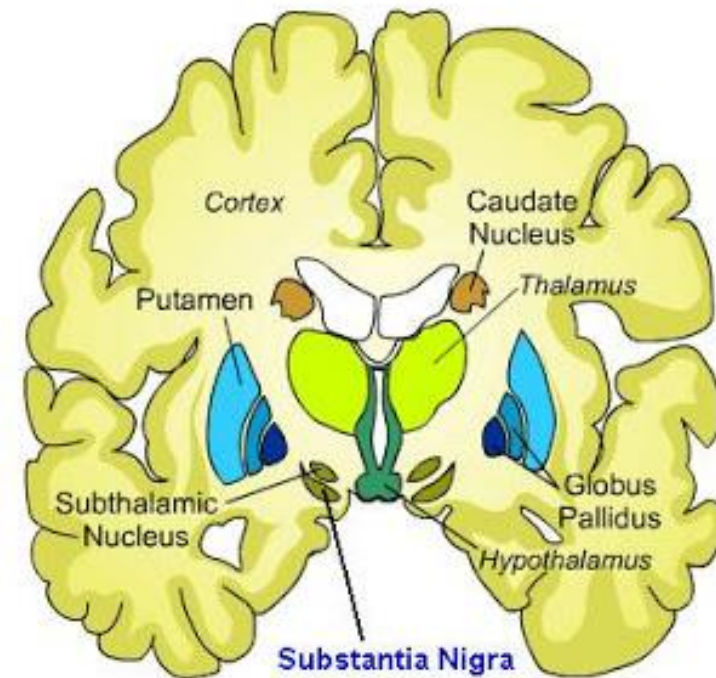
- Lewy bodies:
 - Intracellular inclusion bodies with main constituent being misfolded α -synuclein



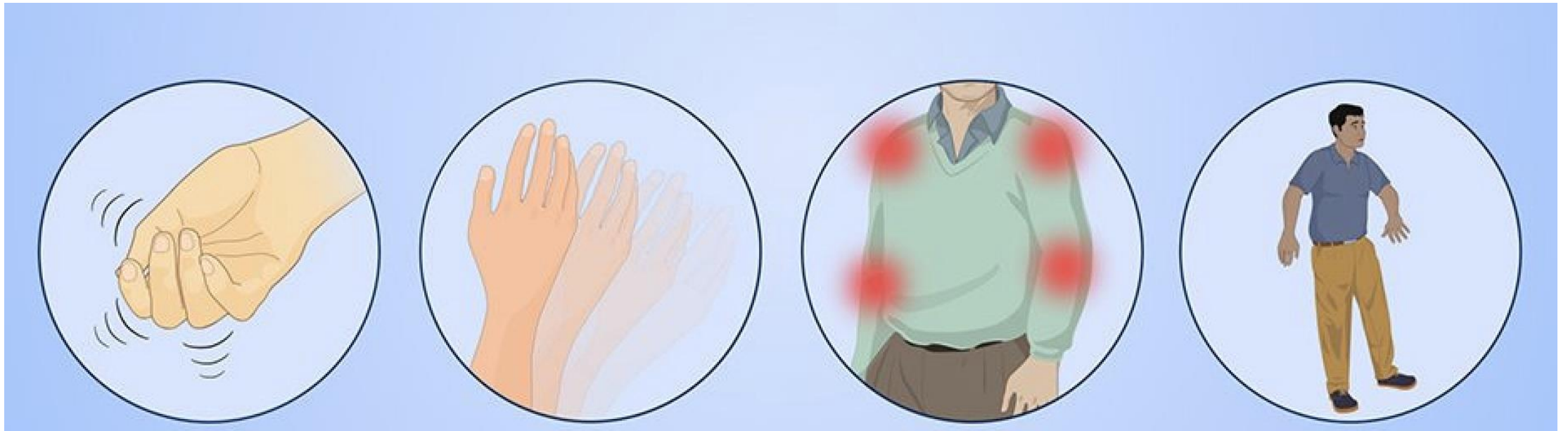
Parkinson's disease: Basal ganglia circuitry



Basal Ganglia: Substantia Nigra, Putamen, Caudate Nucleus, Globus Pallidus



PD: Motor Symptoms



Rest tremor

Bradykinesia

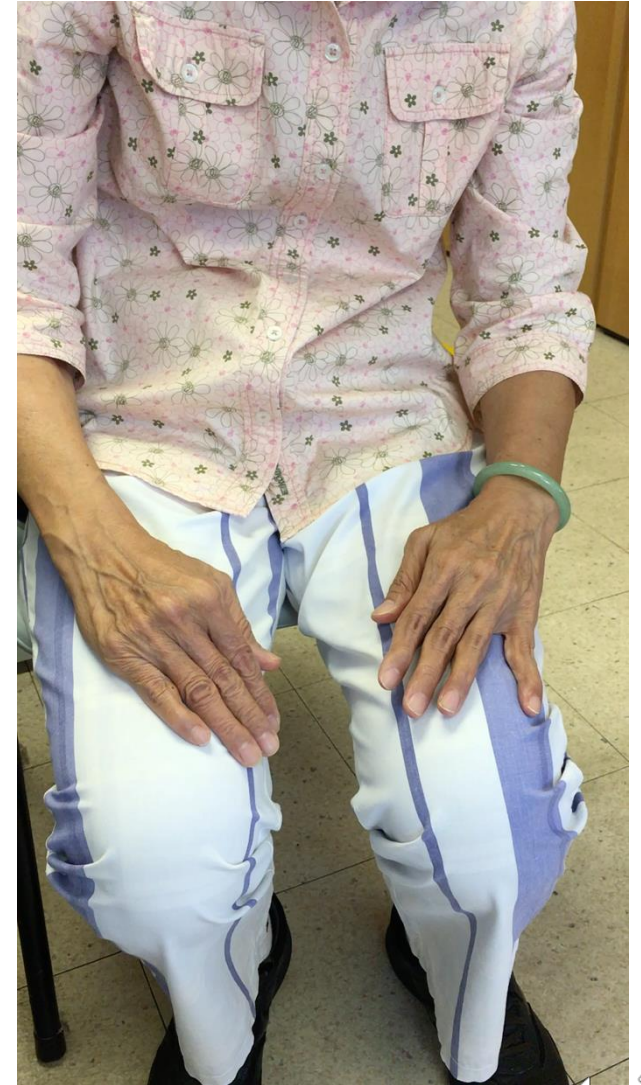
Rigidity

Postural instability



Tremor

- Involuntary, rhythmic, oscillatory movements of part of the body
- Occurs when reciprocal, antagonistic muscles repeatedly contract and relax
- Rest tremor
 - Seen at rest, when a body part is completely supported
- Action tremor
 - Postural: occurs when limb is maintained on one position that is not supported e.g. outstretched arms
 - Kinetic: occurs during movement
 - Simple kinetic tremor: does not change during a movement
 - Intentional tremor: tremor worsens as body part moves towards its target



Causes of tremor

Type of tremor	Causes
Rest tremor	• Parkinson's disease • Parkinsonism
Postural tremor	• Essential tremor • Physiologic tremor (stress, fatigue, sleep deprivation, caffeine) • Drugs (e.g. valproate, theophylline, beta-adrenergic agonists) • Substance withdrawal (e.g. alcohol, sedatives) • Metabolic (e.g. hyperthyroidism, liver failure, hypoglycemia)
Intentional tremor	• Cerebellar disorders



Parkinson's disease: diagnosis

Clinical diagnosis

Presence of motor features:

- Bradykinesia
- Rigidity
- Resting tremor
- Postural instability is typically a feature of more advanced disease



Diagnostic criteria

UK Parkinson's Disease Society
Brain Bank Criteria

MDS Clinical Diagnostic Criteria for
PD

Parkinsonism is defined

Additional features to evaluate
whether parkinsonism is due to PD



UK PD Brain Bank Criteria

Bradykinesia plus one of:

- Muscular rigidity
- 4-6 Hz rest tremor
- Postural instability

Exclusion criteria

Supportive criteria (3 or more)

- Unilateral onset, rest tremor, persistent asymmetry, excellent levodopa response, severe levodopa-induced chorea, levodopa response for 5 years or more, clinical course of 10 years or more



MDS Clinical Diagnostic Criteria for PD

Parkinsonism: bradykinesia plus at least 1 of rest tremor or rigidity

Clinically established PD, clinically probable PD

Supportive and absolute exclusion criteria

Red flags: rapid progression of gait impairment, recurrent falls within 3 y, absence of common nonmotor features despite 5y duration, complete absence of motor progression over 5 or more years, other signs which could suggest atypical parkinsonism

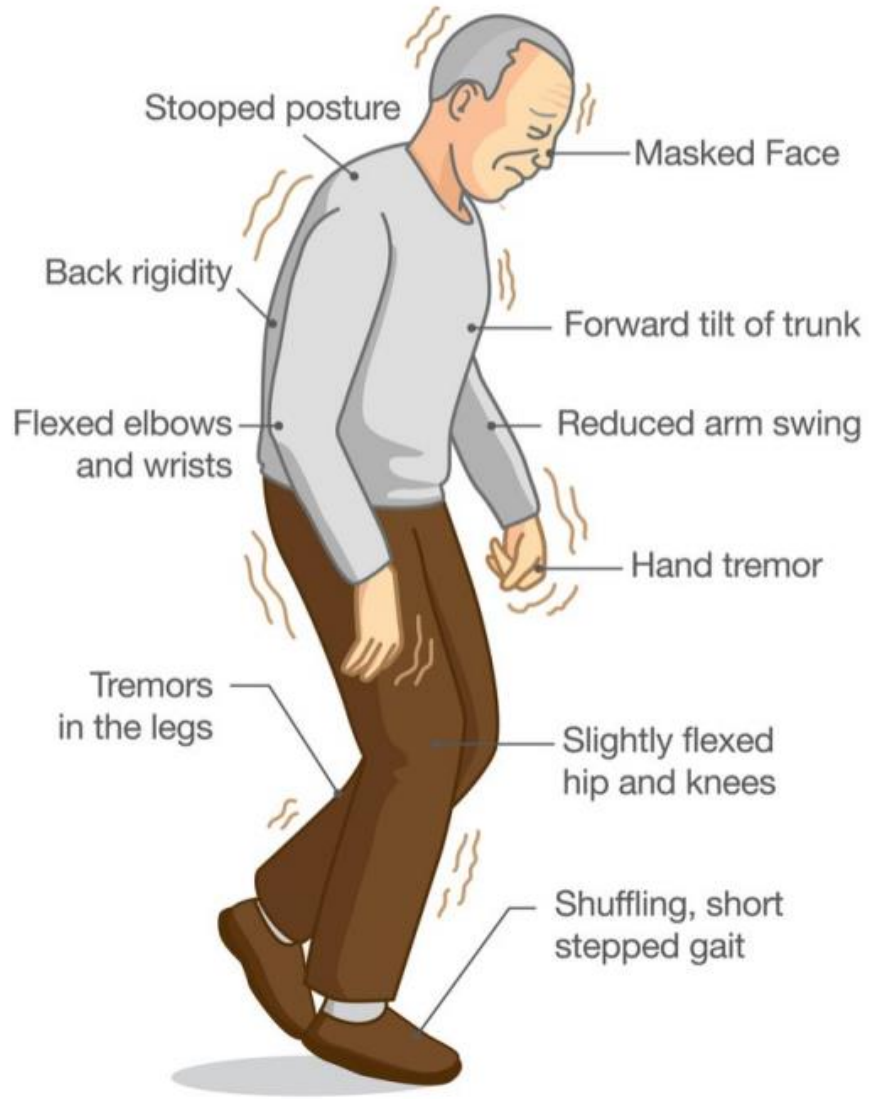


PD features:

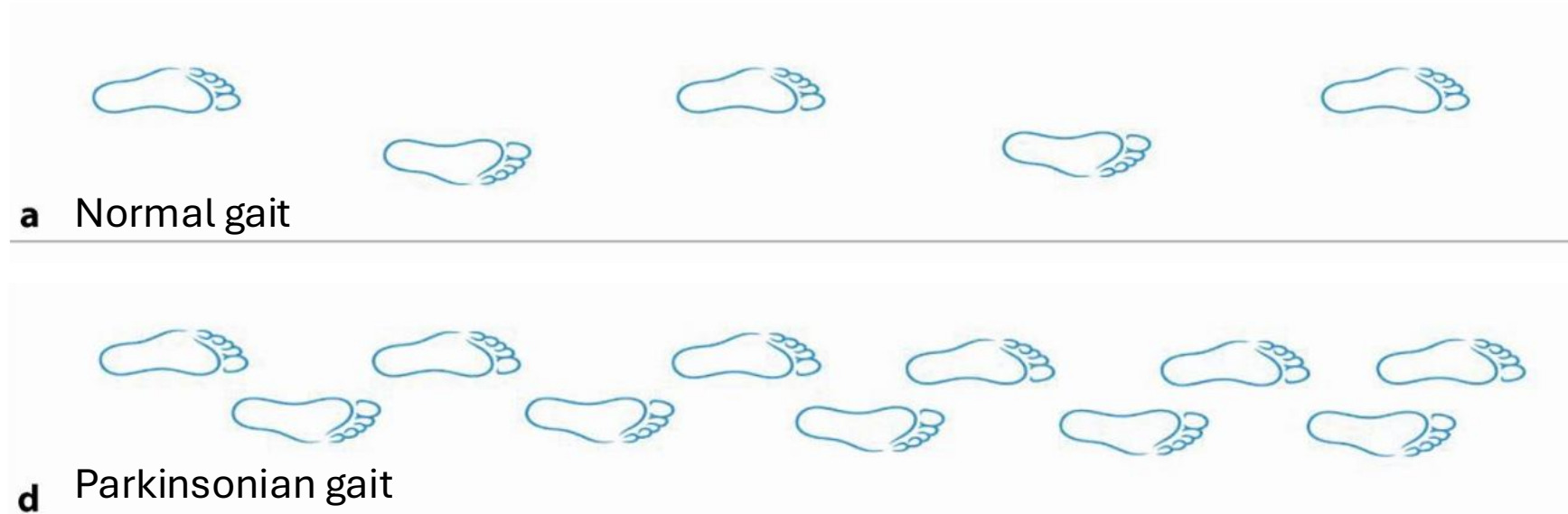
- Rest tremor
- Asymmetry
- Bradykinesia
- Masked face

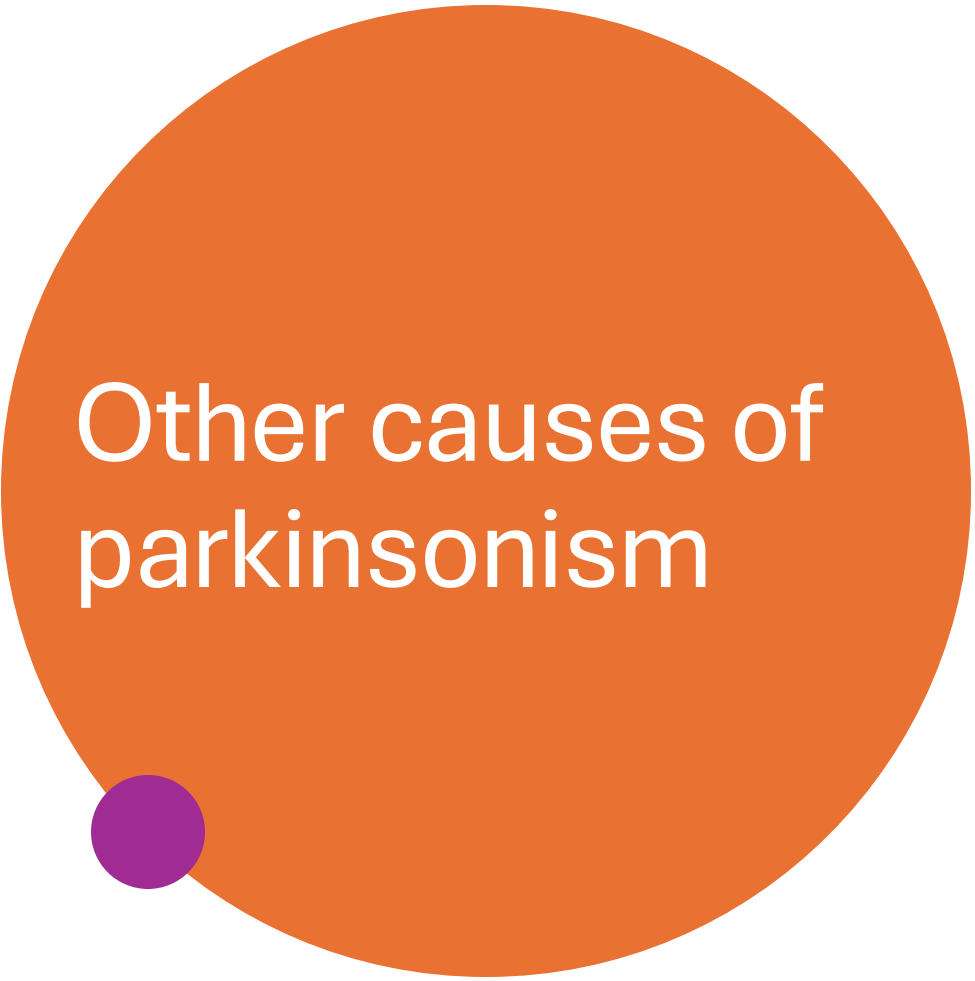


PD: posture and gait



Parkinsonian gait (rigid akinetic gait)





Other causes of parkinsonism

- Atypical parkinsonism or parkinsonism plus syndromes
 - MSA
 - PSP
 - CBD
 - LBD
 - Vascular parkinsonism
 - Drug-induced



PD: Treatment

Drugs

Dopamine
replacement

Surgery

Deep brain
stimulation

Rehabilitation

PT/OT/ST/dietitian



Other Movement Disorders: Chorea, Athetosis and Hemiballismus

Chorea

- Random, flowing, non-suppressible involuntary movements of the distal muscles and face
- May be incorporated into semi-purposeful acts that mask the involuntary movements

Athetosis

- Slow chorea
- Non-rhythmic, slow, writhing movements in distal muscles of hands and feet

Hemiballismus

- Unilateral, rapid, non-rhythmic, non-suppressible, wildly flinging movement of proximal arm and/or leg
- Severe form of chorea



Causes of choreoathetosis

Medication side effects

- Levodopa
- Anti-psychotics

Autoimmune disorder
e.g. SLE

Rheumatic fever

Huntington disease

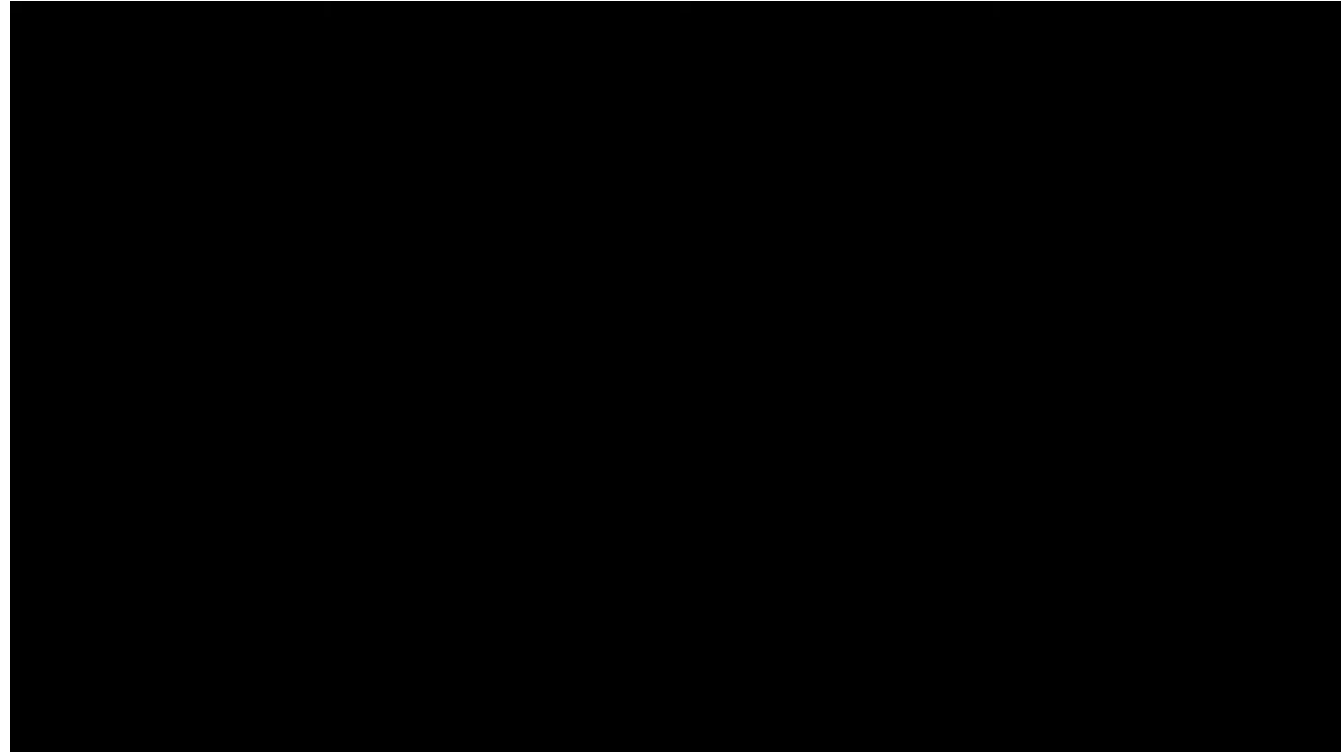
Metabolic:
hyperglycemia,
hyperthyroidism

Stroke:

- Hemiballismus: contralateral subthalamic nucleus
- Hemichorea: contralateral putamen or caudate



Chorea in Huntington's disease



Levodopa-induced dyskinesia



Dystonia

Involuntary muscle contractions which may be sustained or intermittent

May result in abnormal postures, movements, or both

Classification:

- Etiology: primary (inherited, idiopathic) or secondary (CNS disorders or medications)
- Distribution: focal, segmental, generalized



Dystonia: etiology



Primary dystonia

Usually generalized dystonia
beginning in childhood
Most often due to genetic
mutations



CNS disorders

Wilson's disease
Stroke, cerebral palsy
Off symptom of Parkinson's
disease



Drugs

Anti-psychotics
Anti-emetics



Cervical dystonia



Primary generalized dystonia



Dystonia on walking



Myoclonus

- Brief, shock-like, involuntary jerks which result in movement of a joint (positive myoclonus) or with sudden loss of muscle tone (negative myoclonus)
- Physiologic
 - Hypnic jerks
 - Hiccup
- Pathologic
 - Secondary to underlying disorders or medications



Causes of myoclonus

Degeneration of basal ganglia

- Parkinsonian disorders
- Huntington's disease

Dementia

- Creutzfeldt-Jakob disease
- Alzheimer's disease

Encephalopathies and epilepsies

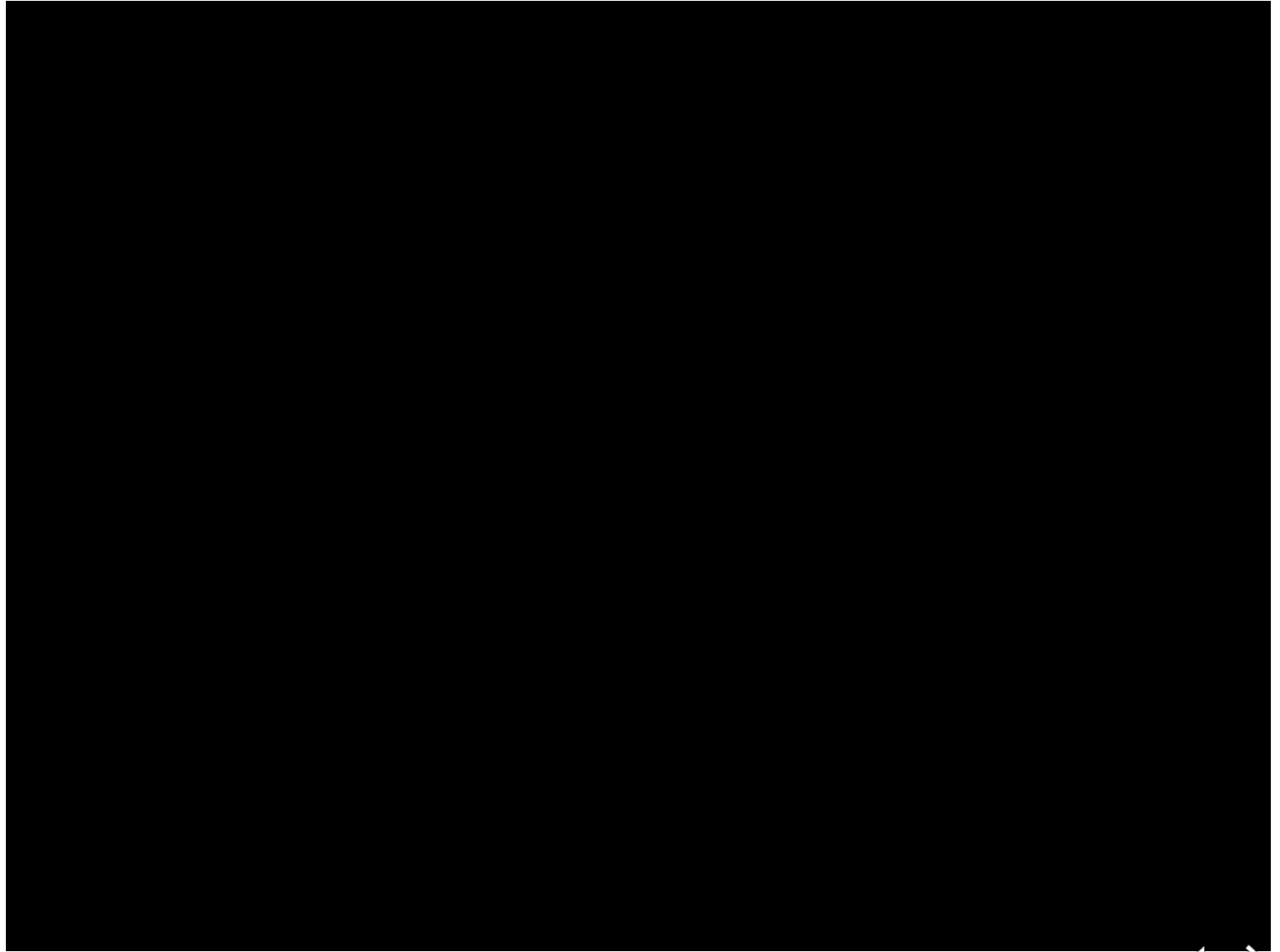
- Mitochondrial disorders, myoclonic epilepsies

Toxic/metabolic causes

- Hypoxic brain injury
- Hypercapnia, hypoglycemia, liver failure, uremia



Myoclonus



Tics

Fast, repetitive movements that result in sudden and difficult to control movements (motor tics) or sounds (vocal tics)

Fairly common in childhood; usually self limiting and improves over time

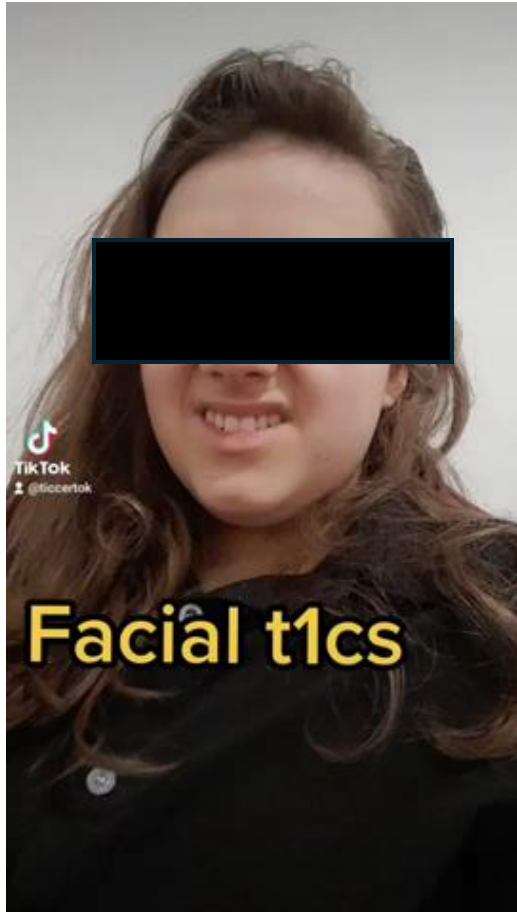
May be associated with other disorders e.g. attention-deficit/hyperactivity disorder, obsessive-compulsive disorder, Huntington's disease

May occur randomly or be associated with triggers

Often start with an unpleasant sensation that builds up until relieved by the tic

Can sometimes be partly suppressed





Examples of tics

- Simple motor tics
 - Blinking, wrinkling the nose or grimacing
 - Shoulder shrugging
- Complex motor ticks
 - Jumping, touching an object
- Simple vocal tics:
 - Coughing, grunting, sniffing
- Complex vocal tics:
 - Repeating own words or phrases, repeating others' words or phrases (echolalia), repeating offensive or obscene words (coprolalia)



Causes of tics

- Tourette syndrome:
 - Presence of both motor and vocal tics that occur everyday or intermittently for at least one year
 - Onset before 18 years of age
 - Exclusion of other causes e.g. medications, other substances or another medical condition
- Insults to basal ganglia
 - stroke, trauma, post-infectious
- Substance/toxin
 - Cocaine, amphetamine
 - Carbon monoxide poisoning
- Developmental disorder
 - Asperger syndrome
- Manifestation of an underlying genetic or chromosomal disorder e.g. neuroacanthocytosis



Take-home messages

Normal gait involves a complex interplay for many parts of the nervous system together with normal musculoskeletal and cardiorespiratory function

Afferent nerves carry sensory information regarding body position sense to a centrally-integrating system consisting of the frontal cortex, brainstem, basal ganglia and cerebellum; movement is then carried out by the motor system of pyramidal tract, neuromuscular junctions and muscles

Lesions in any of these systems will affect gait

In the elderly, impaired gait is usually multifactorial



Take home messages

Cerebellar lesions cause gait ataxia

Parkinson's disease is characterized by rest tremor, bradykinesia, rigidity and in more advanced cases rigid akinetic gait

Other movement disorders include chorea, dystonia, myoclonus and tics

These movement disorders are often symptoms of an underlying disorder



The End

Thank you

